9 Renal Acidification Mechanisms

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CHAPTER OUTLINE

KEY POINTS

- The reabsorption of filtered bicarbonate is critical for acid-base homeostasis because it prevents the loss of a substantial quantity of bicarbonate in the urine. However, it does not generate new bicarbonate, which is necessary to replace the bicarbonate used to buffer endogenous and exogenous acid loads, and therefore is not adequate to maintain acid-base homeostasis.
- Ammonia production and excretion by the kidneys result in equimolar bicarbonate generation and comprise the predominant component of the adaptive changes in net acid excretion that are necessary for acid-base homeostasis.
- Ammonia transport results primarily from specific proteins that enable vectorial and highly regulated transport of either NH_3 or NH_4^+ moieties.
- Titratable acid excretion is determined predominantly by the degree of urine acidification and absolute phosphate excretion. Limits in the acid-base-dependent differences in phosphate excretion are a primary factor limiting titratable acid excretion as a component of net acid excretion.
- Organic acid excretion, while critical for the prevention of nephrolithiasis, is a quantitatively less important component of net acid excretion in humans.

Urine acidification encompasses the separate processes of filtered bicarbonate reabsorption and new bicarbonate generation. Glomerular bicarbonate filtration in a person with a normal glomerular filtration rate (120 mL/min) and serum HCO_3^- (24 mmol/L) is more than 4100 mmol/day. Reabsorption of essentially all filtered bicarbonate is necessary for acid-base homeostasis. Endogenous, and sometimes exogenous, acid production is buffered by bicarbonate, resulting in a need for ongoing renal bicarbonate generation to replace that used for buffering purposes. Normal acid-base homeostasis requires intact functioning of each of these processes.

BICARBONATE REABSORPTION

The proximal tubule reabsorbs ~80% of filtered bicarbonate, the thick ascending limb of the Henle loop reabsorbs ~15%, and the remaining filtered bicarbonate is reabsorbed in distal epithelial segments, beginning in the distal convoluted tubule (DCT) and extending through the collecting duct. In addition, the cortical collecting duct (CCD) can secrete bicarbonate, which is essential in the acid-base response to metabolic alkalosis.

PROXIMAL TUBULE

GENERAL TRANSPORT MECHANISMS

Proximal tubule bicarbonate reabsorption involves several interconnected steps (Fig. 9.1). First, H^+ is secreted into the luminal fluid. Apical Na^+/H^+ exchanger (NHE)-isoform 3 (NHE3) mediates 60% to 70% of H^+ secretion and H^+-ATPase the remainder. In the neonatal kidney, the Na^+/H^+ exchanger, NHE8, substitutes for NHE3.[1] Secreted H^+ combines with luminal HCO_3^- to form carbonic acid (H_2CO_3), which dissociates into H_2O and CO_2. This

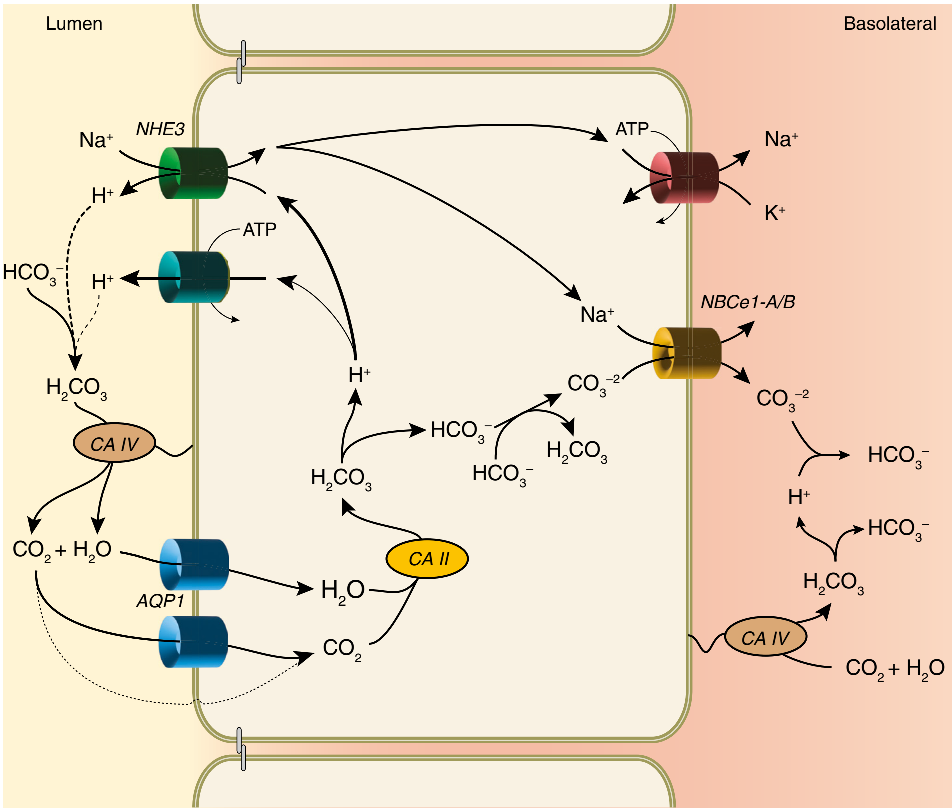


Fig. 9.1 Bicarbonate reabsorption in the proximal tubule. Protons are secreted into the luminal fluid by the Na^+/H^+ exchanger, NHE3, and H^+-ATPase, and titrate luminal HCO_3^- to H_2CO_3. Approximately two-thirds of H^+ secretion is mediated by NHE3 and the remainder by H^+-ATPase. Luminal H_2CO_3 dehydration to H_2O and CO_2 is accelerated by apical CA IV-mediated carbonic anhydrase activity. CO_2 enters the cell via AQP1 and passive lipid-phase diffusion, where cytoplasmic CA II accelerates its hydration to H_2CO_3. H_2CO_3 dissociates to H^+ and HCO_3^-, thereby "providing" cytosolic H^+ for continued apical secretion. Two HCO_3^- convert to H_2CO_3 and CO_3^{2-}, and the CO_3^{2-} exits via basolateral NBCe1 proteins, of which NBCe1-A is the predominant splice variant in the cortex and NBCe1-B is the only splice variant in the PST in the outer medulla. Basolateral CA IV generates H_2CO_3, which dissociates to HCO_3^- and H^+, and the latter converts interstitial CO_3^{2-} to HCO_3^-, completing the process of luminal HCO_3^- reabsorption. Na^+ that entered via apical NHE3 in the process of HCO_3^- reabsorption exits either via basolateral NBCe1-A/B (~75%) or Na^+-K^+-ATPase (~25%). Exit via Na^+-K^+-ATPase maintains the low intracellular Na^+ concentration necessary for continued apical NHE3 activity. In the PST, a basolateral Cl^-/HCO_3^- exchange activity (not shown) is the primary basolateral HCO_3^- exit mechanism.

reaction is catalyzed by carbonic anhydrase IV (CA IV), which is an apical membrane-bound protein. Luminal CO_2 then moves across the apical plasma membrane into the cell through both protein-mediated mechanisms and lipid diffusion.[2] Cytosolic CO_2 is then hydrated to form carbonic acid through a process accelerated by cytosolic carbonic anhydrase II (CA II). Carbonic acid rapidly dissociates to H^+ and HCO_3^-, replenishing the H^+ secreted across the apical plasma membrane. In the S1 and S2 segments, the primary base exit mechanism involves the A splice variant of sodium-coupled, electrogenic bicarbonate cotransporter, NBCe-1A,[3,4] and a secondary mechanism is the B splice variant, NBCe1-B.[5,6] In the S3 segment, a Na^+-dependent, Cl^-/HCO_3^- exchanger is the primary basolateral HCO_3^- transport mechanism.[7]

The proximal tubule also exhibits passive bicarbonate secretion, termed "backleak." Backleak involves both paracellular and transcellular components[8,9] and is inhibited by AngII.[10] Bicarbonate backleak contributes to the proximal tubule's inability to lower luminal bicarbonate below ~6 mmol/L.

REGULATION OF PROXIMAL TUBULE BICARBONATE REABSORPTION

Systemic acid base

Proximal tubule bicarbonate reabsorption responds to acid-base disorders. Both metabolic and respiratory acidosis increase bicarbonate reabsorption, whereas alkalosis has the opposite effect, and the effects are greater with chronic conditions. These effects involve peritubular HCO_3^- and CO_2 signaling through the basolateral membrane protein, protein-tyrosine phosphatase, receptor-type, gamma (PTPRG).[11]

Potassium

Chronic, but not acute, hypokalemia increases proximal tubule bicarbonate reabsorption.[12,13] Increased NHE3 expression has been shown in some[14,15] but not in other[16] studies. In addition, there is increased expression of both NBCe1-A and NBCe1-B[17] and of renal AT1.[18]

Glomerulotubular feedback

Changes in proximal tubule bicarbonate delivery, whether from altered bicarbonate concentration or altered luminal flow rates, induce parallel changes in bicarbonate reabsorption.[19] This is termed "glomerulotubular feedback." Changes in luminal flow rate alter microvillus drag force that is transmitted through the actin filament and alters cytoskeletal elements and transport [(20)].

Angiotensin II (AngII)

AngII has biphasic effects on bicarbonate reabsorption. Low AngII concentrations increase, whereas high concentrations inhibit bicarbonate reabsorption.[21,22] Low concentrations of AngII stimulate bicarbonate reabsorption through activation of both apical and basolateral AT_1 receptors, whereas high concentrations inhibit bicarbonate reabsorption. Acidosis increases AT_1 receptor expression, which may contribute to adaptive changes in bicarbonate reabsorption.[23]

Endothelin

Endothelin can be produced in the proximal tubule and exhibits an autocrine effect to stimulate NHE3.[24] In particular, metabolic acidosis-induced increases in NHE3 expression require ET-B receptor activation.[25]

PTH

PTH acutely leads to metabolic acidosis but chronically leads to metabolic alkalosis.[26] The acute effect is due primarily to increased urinary bicarbonate excretion, as PTH acutely inhibits proximal tubule bicarbonate reabsorption through activation of adenylyl cyclase and increased intracellular cAMP production.[27] The chronic effect is likely mediated by hypercalcemia activating the proximal tubule calcium-sensing receptor (CaSR).

Calcium-sensing receptor (CaSR)

CaSR is present in the apical membrane in the proximal tubule. Its activation increases apical NHE3 activity, leading to increased bicarbonate reabsorption.[28]

PROTEINS INVOLVED IN PROXIMAL TUBULE BICARBONATE REABSORPTION

Na^+/H^+ exchangers

Na^+/H^+ exchanger (NHE) isoform 3 (NHE3 [SLC9A3]) mediates most luminal bicarbonate reabsorption. Several pathways including subcellular distribution, protein-protein interactions, and phosphorylation-dephosphorylation regulate plasma membrane NHE3 activity. Renal sympathetic nerve activity, glucocorticoids, insulin, AngII, dopamine, and PTH are major determinants of the redistribution between apical microvilli and subapical locations.[29] Protein-protein interactions involve the interaction of the C-terminal domain with ezrin, NHERF1, and NHERF2.[30] Phosphorylation and dephosphorylation at several specific residues comprise another major regulatory mechanism.[31,32]

NHE8 (SLC9A8) is a second Na^+/H^+ exchanger found in the proximal tubule.[33] In the adult kidney, NHE8 is mostly intracellular,[29] but in the absence of NHE3 and during acid-loading in normal mice, apical plasma membrane NHE8 protein expression increases.[1] In the neonatal kidney, NHE8 is the predominant proximal tubule NHE isoform.[1,34]

H^+-ATPase

The vacuolar H^+-ATPase mediates ~33% of proximal tubule apical H^+ secretion.[35] This is regulated by AngII, glomerulotubular feedback, and chronic metabolic acidosis.[20,36,37] H^+-ATPase directly interacts with aldolase, which may underlie the development of proximal RTA in individuals with hereditary fructose intolerance.[38] In addition, protein kinase A (PKA) stimulates and the AMP-activated protein kinase (AMPK) inhibits apical plasma membrane H^+-ATPase insertion and activity.[39]

NBCe1 (SLC4A4)

The electroneutral sodium-bicarbonate cotransporter, NBCe1, is responsible for the majority of basolateral bicarbonate exit. Three splice variants are found in humans, but only NBCe1-A and NBCe1-B appear to be expressed in the mammalian kidney.[5] Current evidence suggests NBCe1 proteins transport 1 Na^+ with 1 CO_3^{-2}.[40] Because NBCe1 proteins transport a net negative charge, the intracellular electronegativity provides the electrochemical gradient for transport. Indeed, some forms of proximal RTA result from NBCe1 mutations that result in HCO_3^- transport instead of CO_3^{-2}, which leads to electroneutral transport and thereby decreases net base exit.[41]

NBCe1-A. The A splice variant A of NBCe1 (NBCe1-A) is the primary variant in the kidney. It is found exclusively in the basolateral plasma membrane in the cortical portion of the proximal tubule.[42] Expression increases in response to hypokalemia[17] but not metabolic acidosis.[17,42] Cell signaling pathways that regulate NBCe1-A activity include intracellular ATP, protein kinase C, and calcium/calmodulin-dependent protein kinase II.[43]

NBCe1-B. NBCe1-B expression in the basolateral plasma membrane is found throughout the proximal tubule including in the outer stripe of the outer medulla, where NBCe1-A is not found.[5] In cortical proximal tubule segments, NBCe1-B expression is substantially less than that of NBCe1-A. Metabolic acidosis increases NBCe1-B expression,[5,44] resulting in increased "total" NBCe1 expression[44] and activity[45,46] despite no change in NBCe1-A expression.[17,42]

Carbonic anhydrase

CA II. CA II is the predominant carbonic anhydrase in the kidney. It is located in the proximal tubule cytoplasm of the proximal tubule, in addition to multiple other sites in the kidney, including thin descending limb, TAL, and intercalated cells. In the mouse, CA II is also expressed in collecting duct principal cells.[47]

CA IV. CA IV is found in the proximal tubule and intercalated cells in the collecting duct.[48] CA IV is linked to the plasma membrane via a glycosylphosphatidylinositol anchor and has an extracellular active site.[49] In the proximal tubule, CA IV is expressed in both apical and basolateral plasma membranes.[50]

K^+ channels

K^+ exit via basolateral K^+ channels is necessary for proximal tubule bicarbonate reabsorption. This enables the recycling of K^+ that enters via basolateral Na^+-K^+-ATPase, termed "pump-leak" coupling. Moreover, it allows the coupling of electrogenic basolateral base exit via NBCe1 family members with electrogenic K^+ exit via K^+ channels. Both Kir4.2 and TASK2 K^+ channels are involved in this process.[51] Kir4.2 deletion in rodents causes a proximal renal tubular acidosis phenotype.[51] TASK2 deletion also causes metabolic acidosis.[52] In addition, TASK2 is activated by extracellular alkalinization,[52] which facilitates functional coupling with electrogenic base exit by NBCe1-A and NBCe1-B.

LOOP OF HENLE

The thick ascending limb of the loop of Henle (TAL) reabsorbs ~15% of the filtered bicarbonate load. The overall mechanism is similar to that in the proximal tubule. Apical Na^+/H^+ exchange activity is the primary H^+ secretory mechanism; vacuolar H^+-ATPase activity is present but has only a minor role.[53,54] Two Na^+/H^+ exchanger isoforms are present, NHE2 and NHE3, and NHE3 appears to be the predominant isoform.[55,56] Secreted H^+ reacts with luminal HCO_3^-, forming H_2CO_3, which dissociates to CO_2 and H_2O. Whether luminal CA IV is involved is unclear, with conflicting published findings.[50,57] Luminal CO_2 then enters the cell, where cytoplasmic CA II catalyzes its hydration to form H_2CO_3. This dissociates to H^+ and HCO_3^-, thereby regenerating the H^+ secreted across the apical plasma membrane. Basolateral bicarbonate exit appears to primarily occur through a basolateral Na^+-independent Cl^-/HCO_3^- exchange activity,[58] likely AE2, but a K^+-HCO_3^- cotransport activity mediated by KCC4 may also contribute.[59]

REGULATION OF TAL BICARBONATE REABSORPTION

Whether adaptive changes in TAL bicarbonate transport contribute to acid-base homeostasis is controversial. In one study, NH_4Cl loading (to induce metabolic acidosis), NaCl loading, and $NaHCO_3$ loading each increased TAL bicarbonate reabsorption similarly, suggesting the stimulus was solute loading, not acid-base homeostasis.[60] In another study, NH_4Cl-induced metabolic acidosis, but not NaCl loading, increased TAL NHE3 expression.[61]

Several plasma membrane proteins either directly or indirectly alter bicarbonate reabsorption. Inhibiting the apical Na^+-K^+-$2Cl^-$ cotransporter, NKCC2, such as with a loop diuretic, increases bicarbonate reabsorption.[54] This may reflect decreased NKCC2-mediated apical Na^+ entry decreasing intracellular Na^+, thus increasing the gradient for apical NHE3 activity, and thereby increasing bicarbonate reabsorption. Inhibiting basolateral Na^+/H^+ exchange activity decreases bicarbonate reabsorption through cytoskeletal alterations that decrease apical NHE3 expression.[62]

Several hormones regulate bicarbonate reabsorption. AngII stimulates TAL bicarbonate reabsorption, likely through activation of AT_1 receptors.[63,64] Glucocorticoid receptors are present in the TAL, and glucocorticoid hormones are necessary for basal bicarbonate reabsorption.[65] Mineralocorticoids, at high concentrations, stimulate bicarbonate reabsorption,[66] but their absence does not alter basal transport.[65] Both AVP and PTH inhibit bicarbonate reabsorption through inhibition of apical Na^+/H^+ exchange activity.[67,68] Proinflammatory cytokines regulate bicarbonate transport. Lipopolysaccharide (LPS) inhibits transport through the innate immune system receptor, TLR4. Luminal LPS involves the mTOR pathway, whereas peritubular LPS functions through the MEK/ERK pathway.[69,70]

Medullary osmolality is another crucial regulatory factor. Increased tonicity inhibits and decreased tonicity stimulates bicarbonate reabsorption; this occurs through phosphatidylinositol 3-kinase-mediated changes in apical Na^+/H^+ exchange activity.[71,72]

ACID-BASE TRANSPORTERS IN THE TAL

The major transporters are discussed earlier in relation to the proximal tubule, and they are not repeated here. Please see the relevant section earlier.

DISTAL CONVOLUTED TUBULE

The DCT consists of two cell types, DCT cells and intercalated cells, that have different mechanisms of bicarbonate reabsorption. DCT cells appear to reabsorb bicarbonate through apical H^+ secretion via apical NHE2[56,73] and basolateral HCO_3^- exit via AE2.[74] Cytosolic CA II is present, but not apical CA IV.[50] In the late DCT, A-cells and non-A, non-B cells are present[75] but comprise only a small fraction of the cells in the DCT. Their transport characteristics are similar to intercalated cells in the collecting duct.[76]

COLLECTING DUCT

The collecting duct is the final site of bicarbonate transport. Collecting duct segments contain several distinct cell types, and the cellular composition differs in the various collecting duct segments. In the ICT, CCD, and OMCD, principal cells account for ~60% to 65% of the collecting duct and intercalated cells account for the remainder. A small proportion of cells appear to be transitional cells, with characteristics of both principal cells and intercalated cells.[77] In the IMCD, intercalated cells comprise ~10% of the initial portion of the IMCD and are almost entirely absent by the middle of the papilla. The terminal IMCD is composed of IMCD cells, a cell distinct from both intercalated cells and principal cells.

At least three distinct intercalated cell subtypes exist: the type A (or α) intercalated cell (A-cell), the type B (or β) intercalated cell (B-cell), and the non-A, non-B intercalated cell (non-A, non-B cell) (Fig. 9.2). In the connecting tubule (CNT), both A-cells and non-A, non-B cells are present and B-cells are infrequent. In the CCD, both A-cells and B-cells are present, and the non-A and non-B cells are infrequent. In the OMCD and IMCD, only the A-cell is present.

TYPE A INTERCALATED CELL

The A-cell reabsorbs luminal HCO_3^- (Fig. 9.3). Apical H^+ secretion involves vacuolar H^+-ATPase and the P-type H^+-K^+-ATPase. A Na^+-independent anion exchanger that is a truncated isoform of the erythrocyte anion exchanger, termed *kAE1,* mediates basolateral bicarbonate exit.[78] Cl^- then exits through the combined actions of the KCl cotransporter, KCC4,[79,80] and the basolateral Cl^- channel, ClC-K2.[81,82] Cytoplasmic CA II enables intracellular H^+ and

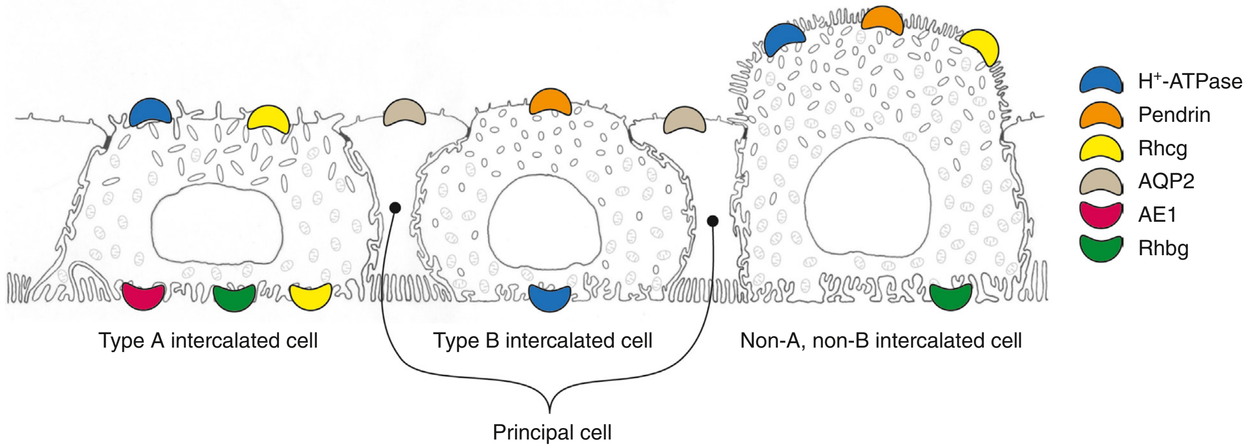


Fig. 9.2 Intercalated cell subtypes in the distal nephron and collecting duct. The late DCT, connecting segment, initial collecting tubule, CCD, OMCD, and IMCD have multiple distinct cell types. Three intercalated cell types can be distinguished on the basis of the differential expression of several proteins involved in renal acid-base transport including H^+-ATPase, AE1, pendrin, Rhbg, and Rhcg. Notably, in the B-cell, H^+-ATPase is present in the basolateral plasma membrane and subapical vesicles but not in the apical plasma membrane. These specific intercalated cell subtypes occur at different frequencies that are specific to the various tubule segments.

HCO_3^- generation. Apical membrane-associated CA IV and basolateral CA XII are present.[83]

TYPE B INTERCALATED CELL

The type B intercalated cell secretes HCO_3^- and reabsorbs Cl^- (Figs. 9.2 and 9.3). H^+-ATPase is present in the basolateral plasma membrane and subapical vesicles but not in the apical plasma membrane, and it secretes H^+ across the basolateral plasma membrane. HCO_3^- is secreted by the apical electroneutral Cl^-/HCO_3^- exchanger, pendrin. Cl^- that enters via pendrin can either recycle across the apical plasma membrane via the apical Cl^- channel, CFTR,[84] or the apical K^+-Cl^- cotransporter, KCC3A,[85] or can exit the cell via the basolateral Cl^- channel ClCK2/Kb.[82] Cytoplasmic CA II facilitates H^+ and HCO_3^- generation. The coupling of pendrin (see later) with these different Cl^- transporters enables various modes of net ion transport. Coupling to CFTR results in electrogenic HCO_3^- secretion. Coupling to ClCK2 results in HCO_3^- secretion with Cl^- reabsorption, which may be beneficial in states of Cl^--depletion metabolic alkalosis. Pendrin may also be coupled to a Na^+-dependent Cl^-/HCO_3^- exchanger (NDCBE), in which two cycles of pendrin-mediated Cl^-/HCO_3^- exchange are coupled to one cycle of NDCBE and result in net NaCl reabsorption but no net acid-base transport.[86]

NON-A, NON-B INTERCALATED CELL

The non-A, non-B cell is present in significant numbers only in the CNT and ICT.[76,87] It is characterized by the apical plasma membrane pendrin and H^+-ATPase expression and the absence of basolateral plasma membrane H^+-ATPase and AE1 (see Fig. 9.2). During development, the non-A, non-B cell arises from different foci than the B-cell, suggesting they are fundamentally different cell types.[88,89]

PRINCIPAL CELLS

Principal cells contribute to bicarbonate reabsorption through indirect and direct roles. Principal cell Na^+ reabsorption leads to luminal electronegativity, which facilitates H^+ secretion by the electrogenic H^+-ATPase. In rodent models, OMCDi principal cells have apical H^+ secretory and basolateral Cl^-/HCO_3^- exchange activities,[90] and they express H^+-ATPase and CA II,[91,92] which suggests they may contribute to bicarbonate reabsorption.

IMCD CELL

The IMCD cell is a distinct cell type and is the predominant cell present in the terminal IMCD. It exhibits carbonic anhydrase activity,[91] both H^+-ATPase and H^+-K^+-ATPase activity,[93,94] and basolateral Cl^-/HCO_3^- exchange activity.[95] However, the expression of each of these is significantly less than in CCD and OMCD intercalated cells and generally cannot be detected by routine immunohistochemistry techniques.

FUNCTIONAL ROLE OF DIFFERENT COLLECTING DUCT SEGMENTS

CNT-ICT

The CNT and ICT contain type A and type B intercalated cell types and non-A, non-B cells, enabling them to contribute to acid-base homeostasis. Under basal conditions, the CNT secretes bicarbonate.[96]

CCD

Unlike the OMCD and IMCD, which only reabsorbs bicarbonate, the CCD simultaneously reabsorbs and secretes bicarbonate. Net transport correlates with systemic acid-base status, with metabolic acidosis stimulating net reabsorption and metabolic alkalosis stimulating net secretion.[97] The ability to secrete bicarbonate, which is not found in the OMCD or IMCD, correlates with the presence of B-cells in the CCD but not in the OMCD or IMCD.

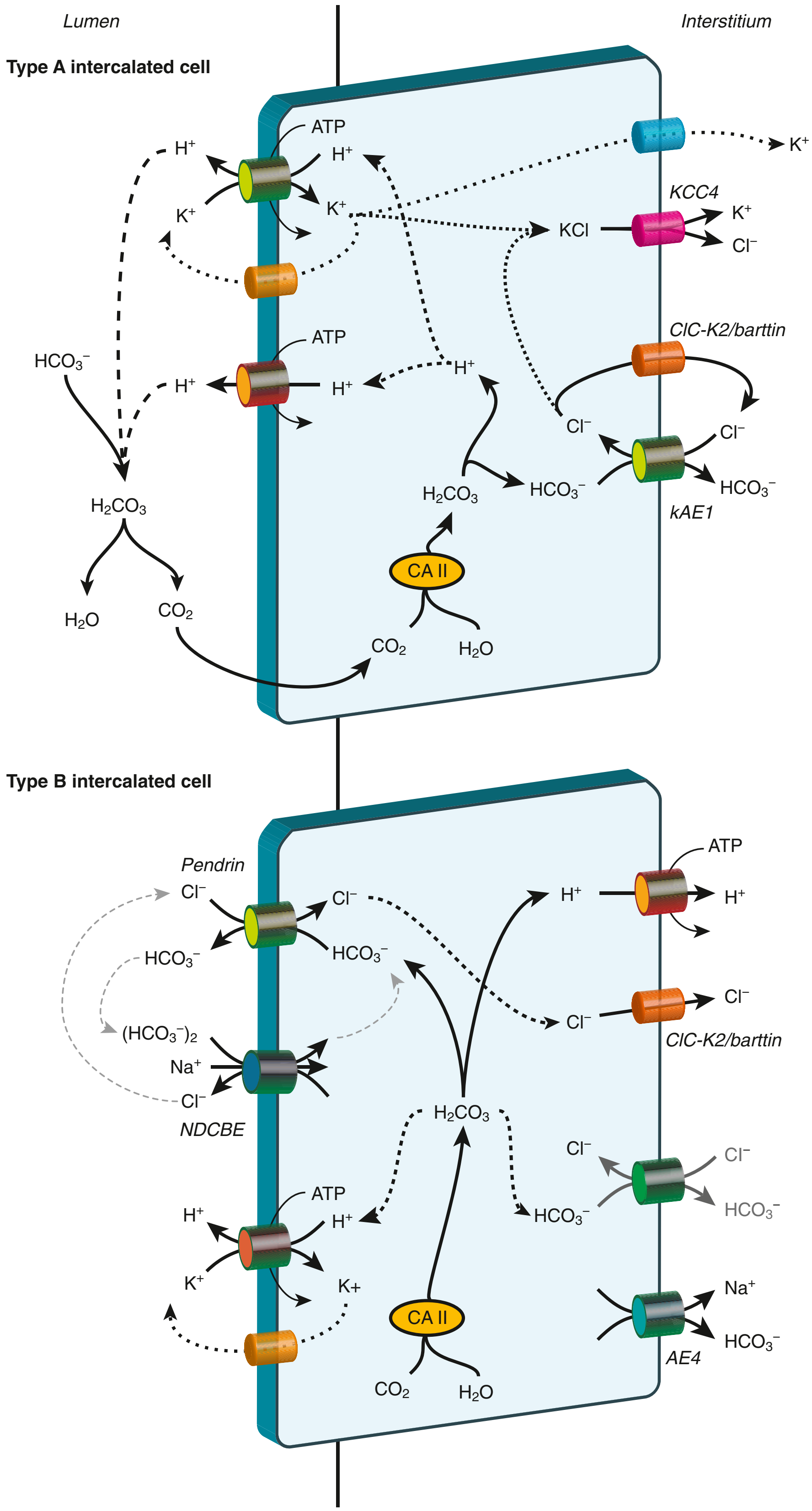


Fig. 9.3 Bicarbonate transport by the A-cell and B-cell. *Top panel,* a model of acid-base transport by the type A intercalated cell. Two families of H^+ transporters, H^+-ATPase and H^+-K^+-ATPase, are present in the apical plasma membrane. Secreted H^+ titrates luminal HCO_3^- to form H_2CO_3, which dehydrates to water (H_2O) and CO_2. Cytosolic H^+ and HCO_3^- are formed from CA II–accelerated hydration of CO_2 to form H_2CO_3, which rapidly dissociates to H^+ and HCO_3^-. Cytosolic HCO_3^- exits across the basolateral plasma membrane via the anion exchanger, kAE1. Cl^- that enters via kAE1 recycles via a basolateral Cl^- channel. K^+ that enters via apical H^+-K^+-ATPase can either recycle via an apical, Ba^+-sensitive K^+ channel or the K-Cl cotransporter, KCC4. *Bottom panel,* H^+/HCO_3^- transport by the type B intercalated cell. Apical pendrin is the primary mechanism of bicarbonate secretion. Chloride enters the cell via pendrin and exits across a basolateral chloride channel, ClC-K2/ClC-Kb, resulting in Cl^- reabsorption coupled to HCO_3^- secretion; recycles via apical CFTR, resulting in HCO_3^- secretion without net Cl^- reabsorption; or couples with apical NDCBE to mediate net NaCl reabsorption without net HCO_3^- transport. Basolateral H^+-ATPase extrudes protons into the peritubular compartment. Cytoplasmic bicarbonate and protons are produced from CO_2 and water in a CA II–catalyzed reaction. In addition, an apical H^+-K^+-ATPase is present. Basolateral AE4 is present and regulates net HCO_3^- secretion via the regulation of apical pendrin.

OMCD

The OMCD is responsible for approximately 40% to 50% of the net acid secretion in the collecting duct. Both intercalated cells and principal cells contribute to acid secretion, but intercalated cells are the primary cells responsible for OMCD acid secretion.[90,98]

IMCD

The IMCD reabsorbs luminal bicarbonate, but absolute rates are substantially lower than in other collecting duct segments.[99] This parallels the smaller number of A-cells in the IMCD.

PROTEINS INVOLVED IN COLLECTING DUCT H^+/ BICARBONATE TRANSPORT

H^+-ATPase

The vacuolar H^+-ATPase is an assembly of multiple subunits that form two main domains: the V_1 domain, which is extramembranous and hydrolyzes ATP, and the V_0 domain, which is transmembranous and transports protons.[100] Distinct isoforms and splice variants have been identified for many of these H^+-ATPase subunits, and their cell-specific distribution may contribute to cell-specific regulation of proton and bicarbonate transport. Normal H^+-ATPase function appears to require the Atp6ap2/(Pro)renin Receptor.[101] This is a type 1 transmembrane protein that is an accessory subunit of H^+-ATPase. Although it is capable of binding prorenin, prorenin binding does not alter H^+-ATPase activity.[102]

H^+-K^+-ATPase

A second mechanism of collecting duct H^+ secretion involves electroneutral H^+-K^+ exchange coupled to ATP hydrolysis.[94] The active protein is a heterodimer composed of α- and β-subunits. The α-subunit is an integral membrane protein with multiple membrane-spanning domains and contains the catalytic portion of the enzyme. Two α-subunit isoforms, HKα1 and HKα2, are present. $HK\alpha_1$ forms heterodimers with its specific β-subunit, HKβ. The β-subunit has only a single membrane-spanning region and is necessary for targeting of the α-subunit to the plasma membrane and for transport function. $HK\alpha_2$ forms heterodimers with the β_1-subunit of Na^+-K^+-ATPase.

Carbonic anhydrase

Three carbonic anhydrase isoforms, CA II, CA IV, and CA XII, are present in the collecting duct. CA II is cytosolic in proximal tubule cells, discussed earlier; in intercalated cells; and in principal cells in mouse collecting duct.[47] CA II is present in all intercalated cell types.

CA IV is an extracellular, membrane-associated carbonic anhydrase tethered to the membrane through a glycosylphosphatidylinositol lipid anchoring protein. It is expressed in the apical plasma membrane in the majority of cells in rabbit OMCD and IMCD and in type A intercalated cells in the CCD.[103]

Carbonic anhydrase XII (CA XII) is a membrane-associated carbonic anhydrase that exhibits basolateral expression in principal cells in the human kidney, but in the mouse kidney, it exhibits basolateral expression in CCD and OMCD A-cells.[83,104]

Anion exchangers

kAE1 (SLC4a1). The primary mechanism of basolateral bicarbonate exit in A-cells is via a truncated form of the erythrocyte anion exchanger AE1, termed kAE1.[105,106] Metabolic acidosis decreases intracellular kAE1 and increases basolateral kAE1, indicating regulated trafficking contributes to bicarbonate reabsorption.[107]

AE4 (SLC4A9). AE4 (Slc4a9) is an electroneutral, cation-dependent, Cl^-/HCO_3^- exchanger. The required cations are either Na^+ or K^+. It is present primarily in the CCD B-cell basolateral plasma membrane, where it regulates the B-cell response to acid-base disorders.[108]

Pendrin (SLC26A4). Pendrin is an electroneutral Cl^-/HCO_3^- exchanger present in the kidney exclusively in the B-cell and non-A, non-B cell. It is found in the apical plasma membrane and apical cytoplasmic vesicles in type B and non-A, non-B intercalated cells in the CNT, ICT, and CCD, and redistribution between these sites is an essential regulatory mechanism.[109] Pendrin is regulated by aldosterone, angiotensin II, nitric oxide, and cAMP.[110,111] In addition to bicarbonate secretion, pendrin is important in extracellular fluid volume and blood pressure regulation. This appears to involve roles in both transcellular Cl^- reabsorption and, through luminal alkalinization due to HCO_3^- secretion, activation of the principal cell epithelial Na^+ transporter, ENaC.[112,113]

NDCBE (SLC4A8). NDCBE mediates the exchange of Na^+-$(HCO_3^-)_2$ with Cl^-. In general, the transmembrane Na^+ gradient enables net Cl^- secretion. NDCBE is expressed in the cortex, where one NDCBE cycle may couple with two cycles of pendrin-mediated Cl^- for HCO_3^- exchange to result in net NaCl uptake.[114] However, NDCBE mRNA expression is substantially less than for pendrin,[115,116] and gene deletion studies show it is not necessary for acid-base or plasma volume regulation in a variety of pathophysiologic conditions.[117]

KCC proteins

KCC3A (SLC12A6). Apical KCC3A is found in the CCD B-cell, where parallel transport by KCC3A and pendrin results in net KHCO3 section.[85] Expression is increased by hyperkalemia, suggesting increased functional coupling of KCC3A, and pendrin may contribute to the metabolic acidosis seen with hyperkalemia.[118]

KCC4 (SLC12A7). Basolateral KCC4 is present in the PCT, TAL, DCT, CNT, and A-cell.[80,119] In the A-cell, KCC4 likely couples with kAE1 to enable basolateral Cl^- recycling. Metabolic acidosis increases OMCD A-cell KCC4 expression,[79] and KCC4 deletion causes distal RTA.[80]

Cl^- channels

ClC-Kb/ClC-K2. Basolateral Cl^- transport via the Cl^- channel, ClC-Kb (in humans, ClC-K2 in rodents), is critical to the function of both type A and type B intercalated cells. In A-cells, basolateral ClC-Kb/K2 is present and likely contributes to recycling the Cl^- that enters via kAE1.[120] In B-cells, basolateral ClC-Kb/K2 likely couples with pendrin to enable transcellular Cl^- reabsorption.[82,120]

CFTR. Apical expression of the Cl^- channel, CFTR, is found in the B-cell, where it enables Cl^- recycling, which facilitates pendrin-mediated bicarbonate secretion.[121] Genetic disorders of CFTR, as occur in cystic fibrosis, impair CCD HCO_3^- secretion in response to metabolic alkalosis.[84]

Sodium-bicarbonate cotransporters

NBCn1 (SLC4A7). The electroneutral sodium-bicarbonate cotransporter, NBCn1 (SLC4a7), is found in the apical region of A-cells in the OMCD.[122,123] It appears to contribute to intracellular pH regulation in these segments[124]; its specific role in transepithelial bicarbonate transport has not been defined. It may also mediate a role in TAL ammonia transport (vide infra).

NBCe2 (SLC4A5). The electrogenic Na^+-HCO_3^- cotransporter isoform 2 (NBCe2, SLC4A5) appears to contribute to acid-base homeostasis. Genetic deletion studies show expression is necessary for normal acid-base homeostasis.[125,126] However, at present, its specific cellular expression is unclear. Different studies have suggested expression in the apical expression in collecting duct intercalated cells in the CCD and OMCD,[127] in microdissected mouse CNT segments,[125] and in the TAL and proximal straight tubule[128] in the rodent kidney. Mouse transcriptomic studies showed NBCe2 mRNA expression predominantly in the OMCD in one study[129] and almost exclusively in OMCD and IMCD1 principal cells in another.[130] Studies examining the human kidney identified expression only in the proximal tubule.[131] Thus NBCe2 appears to contribute to acid-base homeostasis, but its specific role is unclear because of inconsistent findings regarding its cellular expression.

REGULATION OF COLLECTING DUCT ACID-BASE TRANSPORT

The collecting duct is the final site controlling renal acid-base regulation. It responds quickly to physiologic conditions to increase acid or bicarbonate excretion as needed to maintain systemic acid-base homeostasis.

Acidosis

The collecting duct responds to both metabolic and respiratory acidosis with increased HCO_3^- reabsorption. Increased acid secretion in the collecting duct during acidosis is mediated primarily by H^+-ATPase and primarily involves redistribution of H^+-ATPase from a subapical vesicle pool to the apical plasma membrane.[92,132] In most models of metabolic acidosis, total renal H^+-ATPase mRNA and protein expression do not change.[92,133] In parallel, there is increased A-cell AE1, CA II, and CA IV expression.[83,134,135]

Metabolic acidosis decreases CCD unidirectional B-cell HCO_3^- secretion. This results from reduced pendrin expression, resulting in decreased apical Cl^-/HCO_3^- exchange activity in the CCD B-cell.[136-138]

Alkalosis

Metabolic alkalosis induces coordinated changes in acid-base transport throughout the collecting duct. H^+ secretion decreases in the CCD, OMCD, and IMCD.[139,140] In the CCD, bicarbonate loading in animals produces net bicarbonate secretion,[97] which involves increased directional HCO_3^- secretion and decreased H^+ secretion. The intercalated cell response to alkalosis entails essentially the reverse of processes that occur in acidosis.

Potassium disorders (hypokalemia and hyperkalemia)

Potassium disorders commonly cause acid-base disorders. Chronic hypokalemia leads to metabolic alkalosis, likely due to effects on ammonia metabolism.[141,142] In hypokalemia with metabolic alkalosis, CCD and OMCD bicarbonate transport is unchanged.[143] Since alkalosis in the absence of hypokalemia decreases bicarbonate reabsorption, the lack of response in hypokalemic metabolic alkalosis suggests hypokalemia stimulates collecting duct bicarbonate reabsorption. At the cellular level, hypokalemia stimulates A-cells[144-146] and hyperkalemia has the opposite effect.[147] Because in vitro hypokalemia does not alter bicarbonate transport,[148] these effects of K^+ on collecting duct acid-base transport may be indirect.

Hormonal regulation

Hormonal and paracrine factors are important determinants of collecting duct bicarbonate transport. Aldosterone is an important regulator of collecting duct bicarbonate transport.[149] This involves increased H^+-ATPase activity and apical redistribution.[150] Mineralocorticoids also increase CCD bicarbonate secretion; this is dependent on luminal chloride, mediated by pendrin, and involves increased pendrin expression and apical redistribution.[151,152]

Angiotensin II (Ang II) may regulate collecting duct acid-base transport. The collecting duct expresses apical AT1 (AT1a) receptors in both principal cells and intercalated cells.[153] Some studies have shown Ang II increases H^+-ATPase activity in acid-secreting intercalated cells,[153,154] whereas in other studies, Ang II decreased bicarbonate reabsorption in rat OMCD and decreased H^+-ATPase activity.[155,156] This discrepancy is resolved at present.

Endothelin regulates collecting duct acid-base transport. The collecting duct synthesizes endothelin-1 (ET-1),[157,158] and endothelin receptors A and B (ET-A and ET-B) are present in the collecting duct.[159] ET-B activation regulates both A-cell and B-cell responses to metabolic acidosis.[160]

The CaSR is apical in inner medullary collecting duct cells and in type A intercalated cells[161] and mediates luminal Ca^{+2}-stimulation of H^+-ATPase.[162] Luminal acidification stimulated by this pathway may inhibit calcium precipitation, thereby minimizing the development of nephrolithiasis.[162]

Activation of the vasopressin type 1A (V1a) receptor is an additional regulatory mechanism. The V1a receptor is expressed in the mTAL and throughout the collecting duct,[163,164] with expression in both intercalated cells and principal cells in the CCD and intercalated cells in the OMCD.[164] Metabolic acidosis increases V1a receptor expression in the mTAL and OMCD.[164,165] Genetic deficiency of the V1a receptor causes the development of type IV RTA and diminishes mineralocorticoid stimulation of H^+-K^+-ATPase and Rhcg.[166]

The hormone secretin appears to regulate B-cell bicarbonate secretion. Secretin is a gastrointestinal tract–derived hormone that stimulates urine alkalinization through a pendrin- and CFTR-dependent mechanism, and secretin levels increase in response to alkali loading.[84] This response is mediated by the secretin receptor, which is located in the basolateral plasma membrane receptor in B-cells.[167]

Several other hormones and drugs also alter collecting duct acid-base transport. Kallikrein inhibits bicarbonate secretion.[168] Calcitonin stimulates H^+-ATPase-dependent bicarbonate reabsorption in the rabbit CCD.[169] Isoproterenol stimulates B-cell bicarbonate secretion.[170]

The Krebs cycle intermediate, 2-oxoglutarate, regulates collecting duct acid-base transport. Both metabolic acidosis and hypokalemia decrease 2-oxoglutarate excretion.[17,171] In metabolic acidosis, this is determined by changes in transport in the proximal tubule and the loop of Henle.[172] In the CNT and CCD, luminal 2-oxoglutarate, acting through the receptor, Oxgr1, stimulates bicarbonate secretion.[173]

CELLULAR REMODELING

A second adaptive response mechanism used by the collecting duct also involves changes in the number of intercalated cells. Multiple conditions including metabolic acidosis, hypokalemia, and chronic lithium or acetazolamide administration are associated with an increased number of acid-secreting type A intercalated cells and decreased type B intercalated cells.[174-178]

Increases in intercalated cell numbers could result from intercalated cell proliferation or principal cell proliferation, followed by conversion into intercalated cells. Metabolic acidosis, hypokalemia, and lithium administration are all associated with increased proliferation of collecting duct cells.[174,176,179] Some studies increased A-cell proliferation[174,180] while others show the proliferating cells are principal cells.[176,177,179,181]

One major pathway that appears to mediate these changes involves endogenous Notch signaling. Multiple studies show a crucial role for Notch signaling in the interconversion of principal cells and intercalated cells.[182-185] The transcription factors Elf5 and TFCP2L1 appear involved in Notch pathway–dependent principal cell–intercalated cell interconversion.[186,187]

BICARBONATE GENERATION

Acid-base homeostasis requires the generation of new bicarbonate to replace the bicarbonate used to buffer endogenous and exogenous fixed acids. The process of renal bicarbonate generation is also termed "net acid excretion." Clinically, net acid excretion is assessed as urinary titratable acids plus ammonia minus bicarbonate. Quantitatively, changes in ammonia excretion are the predominant component of adaptive changes in net acid excretion (Fig. 9.4).

Because organic anions can be metabolized to form HCO_3^-, their excretion is physiologically equivalent to bicarbonate excretion. Citrate is one such organic anion, and its excretion is decreased by both metabolic acidosis and hypokalemia and increased by metabolic alkalosis. In humans, this response is quantitatively small with respect to acid-base homeostasis but is essential for the prevention of calcium-based nephrolithiasis.[188]

TITRATABLE ACID EXCRETION

Titratable acids are filtered solutes that buffer secreted protons. The secreted H^+ is generated in the cell through the hydration of CO_2 to carbonic acid, with subsequent dissociation to H^+ and HCO_3^-. HCO_3^- is then transported across the basolateral plasma membrane, where it enters peritubular capillaries and is returned via the renal vein, thereby functioning as new bicarbonate generation.

Phosphate is the predominant component, typically accounting for >50% of total titratable acid. Titratable acid excretion in the form of phosphate reflects HPO_4^{-2} that is filtered, not reabsorbed in the proximal tubule, and buffers secreted H^+, forming $H_2PO_4^-$. Filtered $H_2PO_4^-$, which is 20% of total filtered phosphate at pH 7.40, does not buffer secreted $H^+$and therefore does not contribute to titratable acid excretion. Renal phosphate transport is discussed in detail in Chapter 7. Here, we review only the factors that regulate this process in response to acid-base disorders.

Acidosis decreases proximal tubule phosphate reabsorption through several mechanisms, which increases

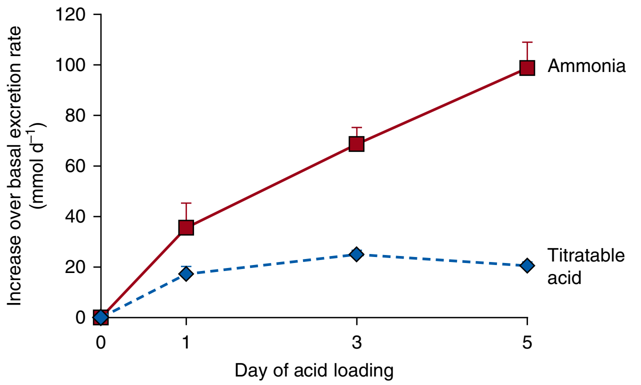


Fig. 9.4 Relative contribution of titratable acid and ammonia excretion in response to metabolic acidosis. Normal human volunteers were acid loaded with ~2 mmol/kg/day of ammonium chloride, and changes in urinary ammonia and titratable acid excretion were quantified. (Data derived from Elkinton JR, Huth EJ, Webster GD Jr, McCance RA. The renal excretion of hydrogen ion in renal tubular acidosis. I. Quantitative assessment of the response to ammonium chloride as an acid load. *Am J Med.* 1960;29:554–575.)

phosphate excretion. First, acidosis decreases NaPi-IIa protein expression, in part due to increased PTH release, and it alters its subcellular distribution, which combine to decrease apical brush border expression.[189] Second, metabolic acidosis stimulates proximal tubule H^+ secretion, which increases the conversion of HPO_4^{2-} to $H_2PO_4^-$. Because $H_2PO_4^-$ is not transported by proximal tubule phosphate transporters, this indirectly decreases total phosphate reabsorption.

Acidosis-induced changes in phosphate excretion depend on adequate phosphate availability. Acidosis increases intestinal phosphate uptake as a result of increased NaPi-IIb expression,[190] and it increases osteoclast-mediated bone phosphate release.[191] When dietary phosphate is restricted, the acidosis-induced changes in renal NaPi-IIa expression and urinary phosphate excretion are blunted.[189,192] In addition, renal NaPi-IIc and Pit-2 expression increase.[192] The net effect of these changes is to limit the development of acidosis-induced hypophosphatemia.

AMMONIA METABOLISM

OVERVIEW OF AMMONIA METABOLISM

Ammonia metabolism, a term used to indicate the combination of ammonia generation and transport, is fundamental to acid-base homeostasis. Ammonia is generated in the kidney through a process that produces an equal amount of bicarbonate. Ammonia transport along the nephron and collecting duct (Fig. 9.5) determines the amount of ammonia excreted in the urine as opposed to entering the peritubular capillaries and being returned via the renal veins. Ammonia added to the renal vein is metabolized in the liver and skeletal muscle in an equimolar bicarbonate-consuming reaction and therefore does not contribute to acid-base homeostasis.

AMMONIA CHEMISTRY

Ammonia exists in two molecular forms, NH_3 and NH_4^+. NH_3, although uncharged, has a significant molecular polarity that limits NH_3 movement across lipid bilayers. NH_4^+ has nearly identical biophysical characteristics as K^+, which enables K^+ transporters also to transport NH_4^+.[193]

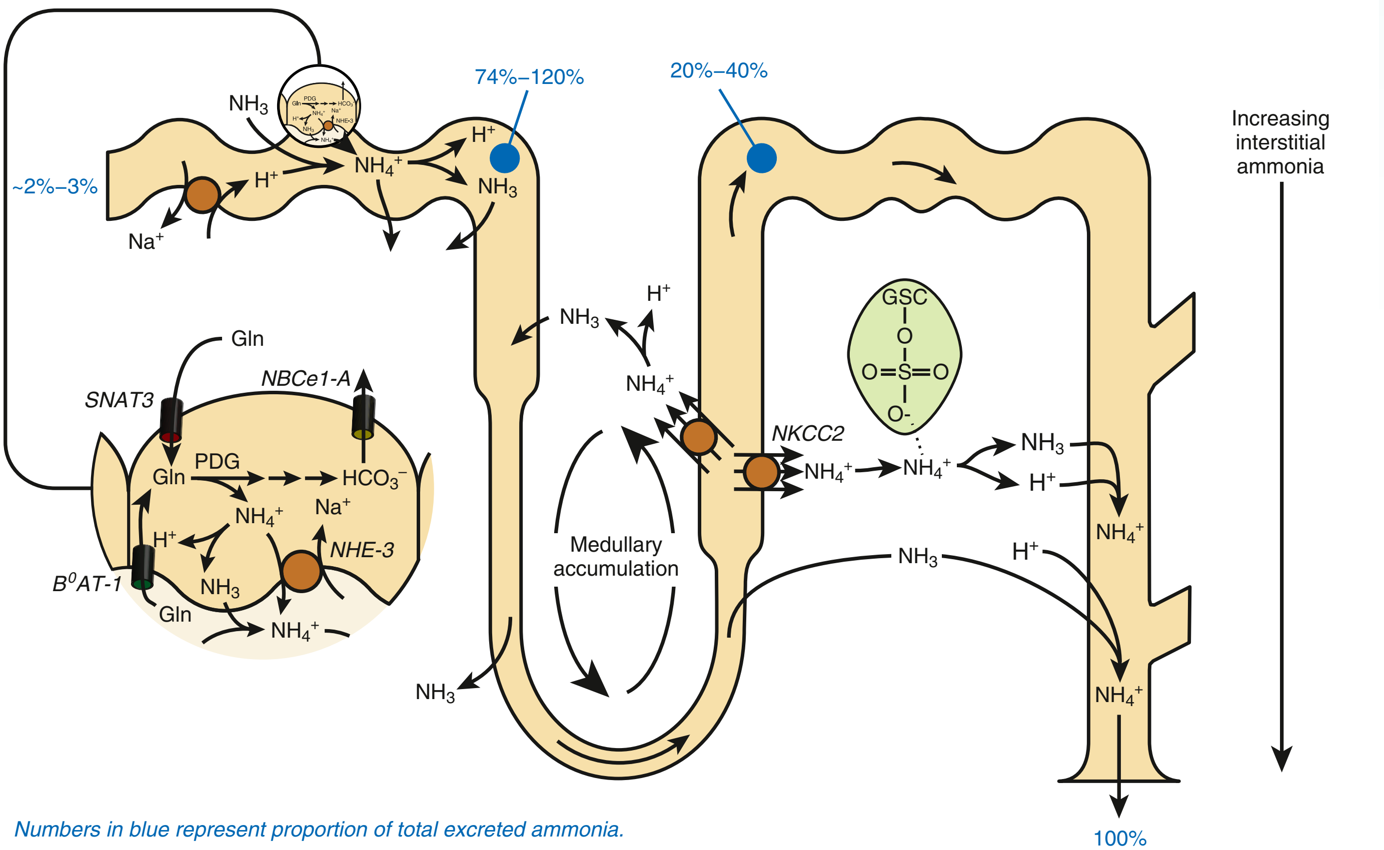
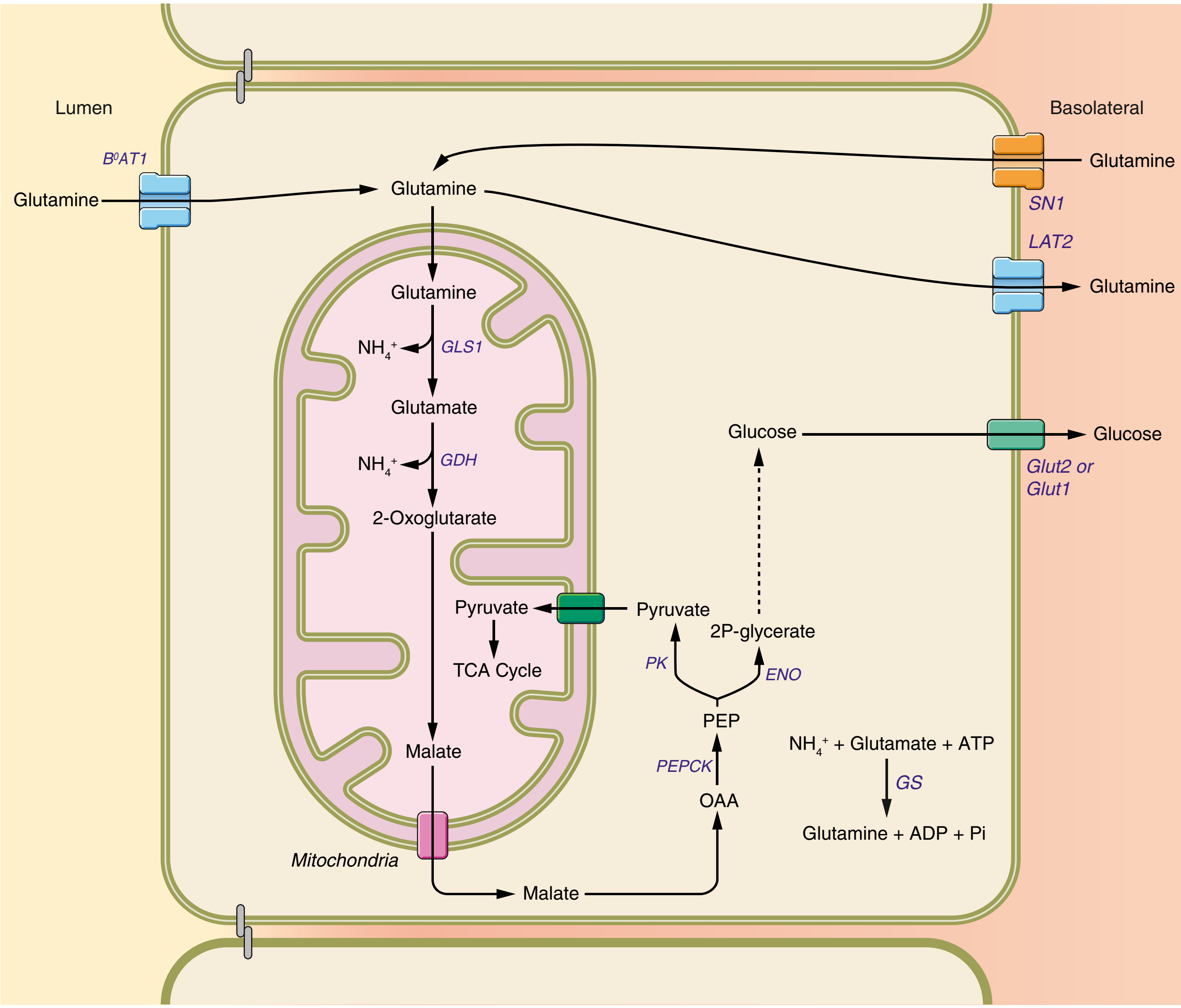


Fig. 9.5 An integrated overview of renal ammonia metabolism. Renal ammoniagenesis occurs primarily in the proximal tubule and generates NH_4^+ and bicarbonate. NH_4^+ is preferentially secreted into the luminal fluid. Ammonia reabsorption in the thick ascending limb, involving apical NKCC2-mediated uptake, results in medullary ammonia accumulation. Medullary sulfatides (highlighted in green) reversibly bind NH_4^+, contributing to medullary accumulation. Ammonia is secreted in the collecting duct via parallel H^+ and NH_3 secretion. The numbers in blue represent the proportion of total excreted ammonia at each location. *Gsc,* Galactosylceramide backbone.

AMMONIA PRODUCTION

Renal ammoniagenesis derives almost exclusively from glutamine metabolism. This process involves both apical and basolateral glutamine transport, mitochondrial glutamine metabolism, cytoplasmic malate metabolism, and differential entry of phosphoenolpyruvate into either the tricarboxylic acid (TCA) cycle or utilization for gluconeogenesis. Fig. 9.6 summarizes these processes.

Glutamine used for ammoniagenesis derives from the uptake of both luminal and peritubular glutamine. The primary apical transporter is B^0AT1 (SLC6A19), with a lesser contribution from B^0AT3 (SLC6A18). Under basal



ATP Generation via TCA and gluconeogenic pathways

TCA pathway

$$2\ \text{Glutamine} + 9\ O_2 \rightarrow 6\ CO_2 + 4\ HCO_3^- + 4\ NH_4^+ + 54\ \text{ATP}$$

Gluconeogenic pathway

$$2\ \text{Glutamine} + 6\ H_2O + 3\ O_2 \rightarrow \text{Glucose} + 4\ HCO_3^- + 4\ NH_4^+ + 14\ \text{ATP}$$

Fig. 9.6 Proximal tubule ammoniagenesis. Glutamine uptake involves apical transport via B°AT1 and basolateral uptake via SN1. Apical glutamine uptake that is not used for ammoniagenesis exits via basolateral LAT2, enabling net glutamine reabsorption under basal conditions. Mitochondrial glutamine is then metabolized by PDG and GDH, yielding 2-oxoglutarate and the release of two NH_4^+. 2-Oxoglutarate is converted to malate in the TCA cycle and then exits the mitochondria. Malate is metabolized to phosphoenolpyruvate (PEP) through a PEPCK-dependent pathway and then either converted to pyruvate via pyruvate kinase (PK) or 2P-glycerate via enolase. Pyruvate enters mitochondria, where its metabolism through the TCA cycle leads to HCO_3^- and ATP generation. 2P-glycerate is metabolized through the gluconeogenic pathway, which also generates HCO_3^- and ATP, but the amount of ATP generated is less than that generated through the pyruvate-dependent TCA cycle.

acid-base conditions, luminal glutamine reabsorption is greater than needed for ammoniagenesis and the remainder exits across the basolateral plasma membrane, leading to net reabsorption. During acidosis, total glutamine uptake can exceed luminal glutamine, and basolateral glutamine uptake mechanisms are stimulated.[194] This basolateral uptake occurs through the Na^+-coupled, neutral amino acid transporter, SN1 (SLC38A3).[195] Basolateral SN1 expression is found only in the S3 segment under basal conditions. In conditions that increase ammoniagenesis, S3 segment expression increases and expression is induced in the S2 segment.[196,197] Glutamine is then transported into mitochondria, where the initial ammoniagenic steps occur.

Glutamine metabolism then leads to the production of 2 NH_4^+ and 2 HCO_3^- molecules from each glutamine. The initial step involves mitochondrial deamidation and deamination by glutaminase (GLS1) and glutamate dehydrogenase (GLDH), respectively, leading to 2-oxoglutarate formation and release of two NH_4^+ from each glutamine. 2-oxoglutarate is metabolized in the TCA cycle to malate, which exits mitochondria via the mitochondrial dicarboxylate transporter, DIC (SLC25A10). Cytoplasmic malate is then converted to phospho*enol*pyruvate (PEP) by phospho*enol*pyruvate carboxykinase (PEPCK). PEP is then a substrate for the TCA cycle and gluconeogenesis. Both pathways generate 2 HCO_3^- molecules per PEP. Because PEPCK is found only in the proximal tubule, glutamine-dependent, ammonia-associated bicarbonate generation is limited to this segment. PEP metabolism through the gluconeogenic pathway enables the kidneys to be an important source of glucose during starvation and during hypokalemia. However, ATP generation is substantially greater through the TCA cycle, ~54 ATP/glutamine, than in the gluconeogenic pathway, ~14 ATP/glutamine. There is also a minor component of ammonia recycling, which involves the conversion of glutamate back to glutamine via the enzyme glutamine synthetase.[198-200]

AMMONIA TRANSPORT

Ammonia produced in the kidney undergoes both reabsorption and secretion in the nephron and collecting duct. In the proximal tubule, it is secreted preferentially into the tubule lumen, which is followed by reabsorption in the TAL. Proximal tubule apical secretion results from NHE3-mediated NH_4^+ secretion and from luminal acidification, which, by decreasing luminal NH_3 concentration, increases the apical gradient for NH_3 secretion.[201,202] The TAL reabsorbs luminal ammonia via the substitution of NH_4^+ for K^+ and reabsorption via the apical Na^+-K^+-$2Cl^-$ cotransporter, NKCC2. Basolateral ammonia exit involves both NHE4-mediated Na^+/NH_4^+ exchange activity[203] and dissociation of NH_4^+ into NH_3 and H^+, with NH_3 transport across the basolateral plasma membrane, and basolateral HCO_3^- entry via NBCn1 that buffers the released H^+.[204] Some ammonia absorbed by the TAL is then secreted by the thin descending limb of the loop of Henle.[205] The net effect of this transport in the proximal tubule and loop of Henle is counter-current amplification and the development of an axial ammonia gradient. The net effect of ammonia transport in the loop of Henle is that ammonia exiting the loop of Henle accounts for only 20% to 40% of urinary ammonia, with the remainder deriving ammonia secretion in more distal segments.

There may be a small component of ammonia transport in the DCT, CNT, and ICT. Studies in rat show low rates of ammonia secretion.[206,207] Ammonia reabsorption may also occur under certain circumstances. Acute increases in ammonia delivery to the DCT and CNT increase blood ammonia levels, suggesting concentration-dependent ammonia reabsorption occurs.[208]

Collecting duct ammonia secretion is responsible for the majority of urinary ammonia. This ammonia secretion involves several transporters (Fig. 9.7). Basolateral uptake in the CCD and OMCD primarily involves the Rhesus glycoproteins, Rhbg and Rhcg,[209,210] although the HCN2 channel may also contribute.[211] In the IMCD,[212] but not the CCD,[213] basolateral Na^+-K^+-ATPase contributes to basolateral NH_4^+ uptake. Apical ammonia secretion involves parallel NH_3 and H^+ transport. H^+ secretion likely involves both H^+-ATPase and H^+-K^+-ATPase. Apical Rhcg mediates the majority of apical NH_3 secretion, and there may also be a smaller component of diffusive transport.[209]

Specific proteins involved in ammonia metabolism

NHE3. The apical Na^+/H^+ exchanger, NHE3, appears to secrete NH_4^+ through the process of Na^+/NH_4^+ exchange.[214] It is likely that hydronium (H_3O^+), not proton (H^+), is the transported moiety in "Na^+/H^+ exchange" and that molecular mimicry of NH_4^+ for H_3O^+ enables Na^+/NH_4^+ exchange. NHE3 is also present in the apical plasma membrane of the TAL. However, NHE3 secretes NH_4^+, whereas the TAL reabsorbs NH_4^+, making NHE3 unlikely to be important in TAL ammonia transport.

NBCe1 family members

NBCe1-A. NBCe1-A is the primary basolateral base transporter in the cortical portion of the proximal tubule. Its regulation of intracellular pH in response to extracellular pH and K^+ changes appears to be the primary mechanism regulating proximal tubule ammoniagenesis under these conditions.[17,42]

NBCe1-B. NBCe1-B appears to regulate the ammoniagenic response of the proximal straight tubule in the outer medulla and to function in parallel with NBCe1-A in cortical proximal tubule segments.[6] Its mechanism of action is likely to be similar to that of NBCe1-A.

Potassium channels

K^+ channels have a variety of roles in ammonia transport and generation. As noted previously, essentially all K^+ channels also transport NH_4^+.[193] In the proximal tubule, K^+ channel inhibition inhibits ammonia transport,[202] but which K^+ channel mediates this is not known currently. In the TAL, K^+ channels contribute to luminal NH_4^+ uptake when apical NKCC2 is inhibited.[215] However, NKCC2 inhibition completely inhibits TAL ammonia transport, indicating apical K^+ channels are unlikely to mediate a significant role in TAL transport.[216]

The K^+ channels, Kir4.2 and Kir5.1, are critical for the regulation of proximal tubule ammoniagenesis. This is likely to occur through voltage-mediated regulation

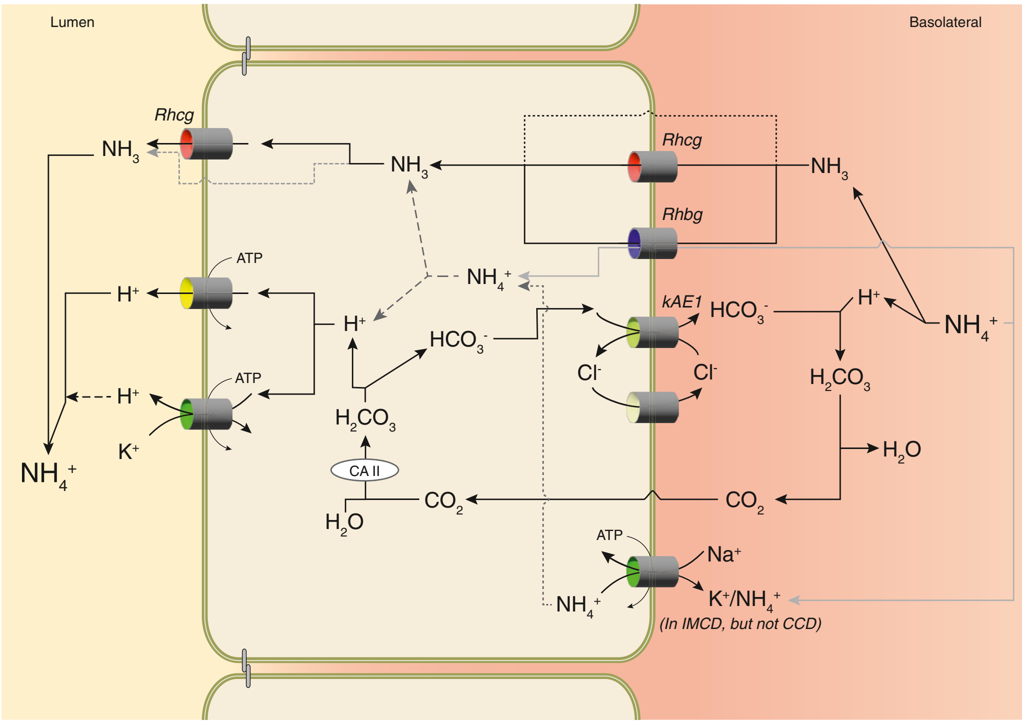


Fig. 9.7 Mechanisms of collecting duct ammonia secretion. Rhbg and Rhcg likely both contribute to basolateral ammonia uptake. Rhbg appears to transport ammonia in both molecular forms, NH_4^+ or NH_3, both of which enable ammonia uptake across the basolateral membrane. Rhcg mediates electroneutral NH_3 uptake across the basolateral plasma membrane. NH_3 is then secreted across the apical plasma membrane down its electrochemical gradient through processes involving both apical Rhcg transport and a separate mechanism that may reflect lipid-phase diffusion. H^+ ions are secreted across the apical plasma membrane by both H^+-ATPase and H^+-K^+-ATPase. Cytoplasmic H^+ are supplied by either dissociation of NH_4^+ or via a carbonic anhydrase II (CA II)-dependent bicarbonate shuttle mechanism involving basolateral chloride-bicarbonate exchange and basolateral chloride channel-mediated bicarbonate shuttling. Also shown is NH_4^+ uptake by basolateral Na^+-K^+-ATPase, which contributes to ammonia secretion in the IMCD, where most cells do not express Rh glycoproteins.

of basolateral NBCe1-mediated base exit, with indirect effects on intracellular pH, which then regulates ammonia generation.

Na^+-K^+-$2Cl^-$ Cotransport

NKCC2. NKCC2 (SLC12A1) is a Na^+-K^+-$2Cl^-$ cotransporter expressed in the apical plasma membrane of the TAL, where it is the primary mechanism for luminal ammonia uptake.[216] The molecular similarity of NH_4^+ and K^+ enables luminal NH_4^+ to compete for transport with K^+. Consequently, in hypokalemia and hyperkalemia, changes in luminal K^+ alter NH_4^+ reabsorption and lead to alterations in medullary interstitial ammonia concentration.

Na^+-K^+-ATPase. In the IMCD, Na^+-K^+-ATPase-mediated basolateral NH_4^+ uptake is critical for IMCD ammonia and acid secretion.[217,218] Furthermore, decreased interstitial K^+ during hypokalemia increases basolateral NH_4^+ uptake, which leads to increased NH_4^+ secretion.[219] This role of Na^+-K^+-ATPase may be specific to the IMCD as Na^+-K^+-ATPase does not contribute to CCD ammonia secretion.[213]

H^+-K^+-ATPase. Potassium deficiency increases expression of the colonic H^+-K^+-ATPase, and it has been postulated that this mediates increased NH_4^+ secretion via transport at the H^+ binding site.[220] This may involve molecular mimicry of NH_4^+ with hydronium (H_3O^+).

Aquaporins

AQP8 may contribute to renal ammonia metabolism. In cultured proximal tubule cells, AQP8 is located in the inner mitochondrial membrane and its knock-down decreases ammonia secretion.[221] Consistent with a role in regulated ammonia generation, metabolic acidosis increases AQP8 expression in vivo.[221] However, mice with AQP8 gene deletion have normal ammonia excretion under basal conditions and following acid loading.[222]

Carbonic anhydrase

Carbonic anhydrase, in addition to its role in bicarbonate reabsorption, also contributes to ammonia secretion. Direct studies have shown that carbonic anhydrase inhibition, presumably through effects on CA II, blocks OMCD ammonia secretion.[223] Apical plasma membrane CA IV likely decreases collecting duct ammonia secretion because it prevents the formation of a luminal disequilibrium pH. Apical CA IV expression has been demonstrated in the rabbit CCD type A intercalated cell, in the rabbit OMCD and IMCD,[103] in the human CCD and OMCD,[224] but not in the rat collecting duct.[50]

Rh glycoproteins

Rhbg (SLC42A2). Rhbg exhibits exclusively basolateral expression in the kidney and is found solely in distal epithelial segments.[47,225,226] A-cells' non-A, non-B cells' principal cells; and connecting segment cells express Rhbg, with expression greatest in intercalated cells. Both metabolic acidosis and hypokalemia increase Rhbg expression, and genetic deletion of Rhbg impairs ammonia excretion.[227,228] Rhbg appears to transport both NH_3 and NH_4^+, but the relative magnitude of NH_3 versus NH_4^+ transport is not well understood.[209] Notably, the electrochemical gradients present in the collecting duct favor Rhbg mediating cellular uptake of both forms.

RhCG/Rhcg (SLC42A2). Rhcg expression essentially parallels Rhbg expression, with the exception that Rhcg exhibits both apical and basolateral expression.[229-231] Rhcg transports ammonia almost entirely in the form of NH_3.[209] This molecular specificity is necessary for apical Rhcg to secrete ammonia. Otherwise, intracellular electronegativity would favor electrogenic NH_4^+ reabsorption across the apical plasma membrane.

Rhcg also regulates collecting duct ammonia secretion through effects on H^+-ATPase. Rhcg and H^+-ATPase are located within the same cellular protein complex, and their interaction modulates H^+-ATPase activity and expression.[232]

Sulfatides

Interstitial sulfatides are important in renal ammonia metabolism. They reversibly bind interstitial NH_4^+, which facilitates the axial ammonia gradient that is necessary for collecting duct ammonia secretion. Their levels are highest in the outer and inner medulla, and metabolic acidosis increases their expression acidosis.[233]

ORGANIC ANION EXCRETION

Urinary organic anions, of which citrate is the primary component, contribute to acid-base homeostasis because their metabolism otherwise generates bicarbonate. Thus their excretion is equivalent to alkali excretion. Citrate is the urinary organic anion found in the largest concentration. Both metabolic acidosis and hypokalemia decrease citrate excretion,[234] as do carbonic anhydrase inhibition and high dietary NaCl or protein intake.[235,236] Although therapeutic lithium chloride doses alter citrate excretion in murine models,[237] this does not occur in humans.[238]

Proximal tubule citrate transport is the primary determinant of citrate excretion. Apical citrate transport is mediated primarily by the sodium-dicarboxylate cotransporter NaDC1.[239] Citrate transported into proximal tubule cells is fully metabolized in the TCA cycle, where it accounts for ~15% of proximal tubule oxidative metabolism, as well as a bicarbonate source.[235]

Multiple mechanisms regulate proximal tubule citrate reabsorption. The primary mechanism appears to be the expression of NaDC-1, which is increased by both metabolic acidosis and hypokalemia,[240,241] and this seems to be determined by intracellular pH.[241] Another mechanism results because NaDC1 transports citrate only in the molecular citrate^{-2}, not as citrate^{-3}. Luminal acidification, as occurs with metabolic acidosis and hypokalemia, converts citrate^{-3}, which is not transported, to citrate^{-2}, which is transported. This change in the molecular citrate moiety enables changes in transport independent of changes in NaDC1 expression.

DIURNAL VARIATION IN ACID EXCRETION

There is a diurnal variation in ammonia and titratable acid excretion and urine pH.[242] Many, but not all, genes encoding the essential proteins involved in bicarbonate reabsorption and net acid excretion either exhibit circadian variation in expression levels or have their expression regulated by circadian transcription factors.[243,244] This diurnal variation is altered in uric acid stone formers, which may contribute to stone development.[245]

SEX DIFFERENCES

Sex differences between males and females have been identified in essentially every organ in the body, and the kidney is no different. The male kidney is larger than the female kidney, but the female kidney, despite being smaller, has twofold more net acid excretion that results from increased ammonia excretion.[246] Several of the critical proteins in ammonia metabolism, including PEPCK, NKCC2, Rhbg, and Rhcg, also exhibit greater expression in the female kidney.[246] Both the larger size and lesser ammonia excretion by the male kidney are primarily dependent on testosterone acting through the androgen receptor.[247,248]

In humans, urine pH is more alkaline in adult women than in adult men.[249] This appears to result from higher rates of intestinal organic anion absorption, leading to an alkaline load from the metabolism of the organic anions, which then stimulates the urine alkalinization that is observed.[249]

ACID-BASE SENSORS

Acid-base and several electrolyte disorders are known to alter net acid excretion, as discussed earlier, and specific mechanisms to transduce these extracellular signals. Several proteins involved in this process are discussed later.

ACID/ALKALI-SENSING RECEPTORS

GPR4

GPR4 is a member of the G protein–coupled receptor family and was one of the first proteins identified as altering renal net acid excretion. Its deletion results in mild metabolic acidosis, more alkaline urine, and decreased ability to excrete an acid load.[250] In acid-loaded mice, GPR4 appears to regulate both intercalated cell AE1 expression and collecting duct cell type remodeling responses.[251] Which renal cells express GPR4 has not been explicitly determined. Transcriptomic studies examining the mouse kidney suggest low expression in renal epithelial cells,[129,130] and in the human kidney, transcriptomic studies suggest expression is limited to endothelial cells, fibroblasts, and lymphocytes.[252]

INSULIN RECEPTOR–RELATED RECEPTOR

Insulin receptor–related receptor (InsR-RR) is expressed in the basolateral plasma membrane of the B-cell and non-A, non-B intercalated cells, where it regulates pendrin expression and pendrin-mediated bicarbonate secretion.[253] InsR-RR is activated by alkaline pH,[254] and its expression is necessary for the expected response to an alkali load.[254]

KINASES

PYK2

Pyk2 is a nonreceptor tyrosine kinase with multiple roles in renal acid-base homeostasis. In cultured proximal tubule cells, Pyk2 is activated by extracellular acidosis and through c-Src activation leads to NHE3 activation.[255,256] In cultured OMCD cells, Pyk2 is required for acidosis-induced activation of H^+-ATPase through a signaling mechanism that involves the MAPK signaling pathway ERK1/2.[257] In mice, Pyk2 deletion partially blunts adaptive changes in citrate transport and excretion in response to acid loading but does not alter serum bicarbonate either under basal conditions or after acid loading.[258]

RECEPTOR TYROSINE PHOSPHATASE

HCO_3^- and CO_2 regulate proximal tubule HCO_3^- reabsorption through receptor protein tyrosine phosphatase-γ (RPTPG) activation.[11] RPTPG has an extracellular carbonic anhydrase-like domain that lacks the critical histidine residues necessary for carbonic anhydrase catalytic activity, but it does allow CO_2 and HCO_3^- binding, which then alters phosphatase activity.[259] Mice with RPTPG deletion exhibit spontaneous metabolic acidosis with decreased ability to respond to acid loading, and their proximal tubules have reduced ability to alter bicarbonate reabsorption in response to pCO_2 or HCO_3^-.[11]

BICARBONATE-STIMULATED ADENYLYL CYCLASE

Cytosolic pH changes may alter renal epithelial cell functioning through the soluble adenylyl cyclase (sAC). sAC is expressed in the TAL, DCT, and collecting duct, and its cAMP production is stimulated by cytoplasmic HCO_3^-.[260,261] In collecting duct intercalated cells, sAC colocalizes with H^+-ATPase in both the A-cell and B-cell, and it coimmunoprecipitates with H^+-ATPase.[261] In clear cells of the epididymis, a model system of collecting duct–intercalated cells, sAC regulates apical H^+-ATPase expression through changes in cAMP production.[260] Changes in cAMP levels also alter the expression of pendrin protein in type B and non-A, non-B intercalated cells in cultured CCD and CNT.[111]

ION TRANSPORTERS

NBCE1 SPLICE VARIANTS

NBCe1-A

The A splice variant of NBCe1, NBCe1-A, is both the primary mechanism of basolateral bicarbonate exit in cortical portions of the proximal tubule and a significant regulator of proximal tubule ammonia and citrate handling. NBCe1-A deletion results in severe metabolic acidosis yet inhibits both basal and stimulated proximal tubule basal ammonia generation capacity and ammoniagenic responses.[17,42] This likely reflects impaired base exit causing intracellular alkalinization, which then inhibits ammoniagenic responses. Hypokalemia appears to increase proximal tubule ammoniagenesis by hyperpolarizing the basolateral plasma membrane, which increases electrogenic base exit via NBCe1-A and leads to intracellular acidification. NBCe1-A also regulates proximal tubule citrate transport and NaDC1 expression through similar effects on intracellular pH.[241]

NBCe1-B

NBCe1-B, at least in the absence of NBCe1-A, regulates the proximal tubule ammoniagenic protein expression both under basal conditions and in response to metabolic acidosis.[6] This effect is most notable in the proximal straight tubule in the outer medulla but is also present in cortical portions of the proximal tubule. The mechanism of action is likely similar to that used by NBCe1-A.

Kir4.2 (KCNJ15)

Kir4.2 is a proximal tubule basolateral K^+ channel that is an important determinant of basolateral plasma membrane voltage. Because membrane voltage regulates NBCe1-A/B base exit, Kir4.2 regulates intracellular pH. Its deletion leads to spontaneous metabolic acidosis and impairs the ammonia excretion response to acid loading.[51]

AE4 (SLC4A9)

AE4 is an electroneutral anion exchanger present in the basolateral plasma membrane of intercalated cells.[108] It mediates the exchange of X^+-$(HCO_3^-)_2$ for Cl^-, where X^+ can be either Na^+ or K^+.[262] Through this HCO_3^- transport activity, it enables extracellular HCO_3^- changes to alter intracellular pH. AE4 deletion impairs acid-base homeostasis by blunting changes in pendrin expression and apical HCO_3^- secretion in response to acid-base disorder.[108]

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